

Thank you for purchasing a FANUC Robot.

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- Straight-head screwdriver
- Torque wrench
- Cross-head screwdriver
- Metric hex-head wrenches

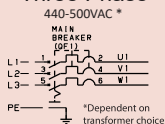
Required Items

- Properly rated transport equipment as specified
- Customer-supplied properly rated power source as specified on the Controller Power Source Label
- Customer-supplied properly rated input power cable; conductor size of AWG10; terminal size is M5
- Multimeter
- Rigid platform to hold robot
- 4-M16 x 65mm bolts
- LOCTITE #243
- 4-M16 high strength washers

Optional Items

- Strain relief cable grip
- Robot locator pins

Three Phase



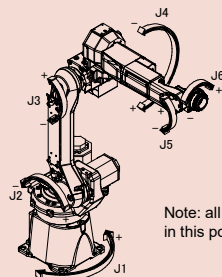
Controller

- Breaker (Also called Circuit Breaker, Main Breaker, or QF1)
- Latch
- F Number and Power Requirements Label



Operator Panel

Robot (mechanical unit) with joints indicated



Note: all axes are 0° in this position

M-10iA variant shown

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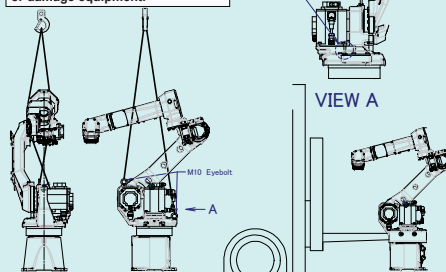
A. Unpack and Install your Robot

1. Cut the base of the plastic covering all the way around the robot and controller. Remove and discard the plastic.
2. Remove the bolts holding the robot to the skid.
3. Use a properly rated fork lift or crane to move the robot to its new location.

WARNING

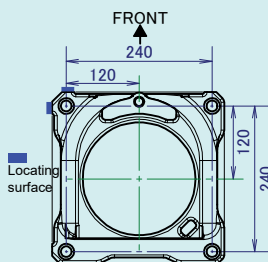
Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

Requirements:
 Crane capacity min: 300kg
 Sling capacity min: 250kg
 Forklift capacity min: 300kg

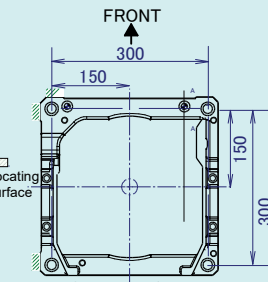


ARC Mate 100iC, M-10iA shown.

4. Fasten the robot base to the floor plate with 4-M16 washers and 4-M16 x 35 bolts (4-M16 x 40 bolts, for ARC Mate 100iD, M-10iD and ARC Mate 120iC, M-20iA) (tensile strength 1200N/mm² or more). Tighten them to a torque spec of 318Nm. If the highest accuracy is desired, locator pins can be used to ensure the robot and a future replacement are mounted in the same location. More information is available in the ARC Mate 100iC / M-10iA or ARC Mate 100iD / M-10iD Mechanical Unit Operator's Manual.

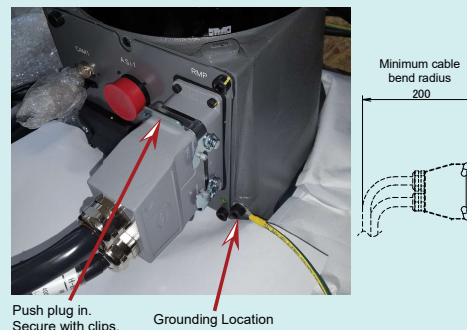


ARC Mate 100iC, M-10iA



ARC Mate 100iD/M-10iD, ARC Mate 120iC/M-20iA

5. Connect the ground wire from the controller to the back corner of the robot base as shown below.
6. Connect the RMP cable to the back of the base. This cable is also referred to as the Robot Connection Cable (RCC).



7. Connect the teach pendant to its cable as shown below.



Robot installation is complete. You are now ready to go to **B. Unpack and Install your Controller.**

B. Unpack and Install your Controller

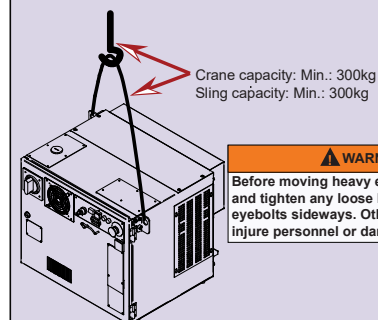
1. Open all of the boxes included in the shipment.



Robot, Controller, and Accessory Box



2. Use properly rated equipment to lift the controller.



3. Place the controller in a safe location where the grounding wire and RMP cable can reach the robot, and the controller is safely reachable from outside of the robot work area.

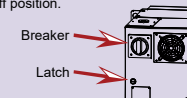


4. Determine the power requirements for your controller from the label on the front. The label is at the bottom left corner of the controller door.

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

5. Open the controller door by using a straight-head screwdriver to turn the latch clockwise, then turn the Breaker counter-clockwise past its off position.



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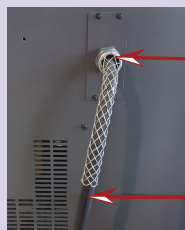
MARGUM10A04171E REV D

6. Unscrew the round white plug from the side of the controller. Remove the four screws highlighted in the illustration below, and carefully remove the breaker access panel. It may stick as it has a sealant around the edges.



Unscrew marked screws

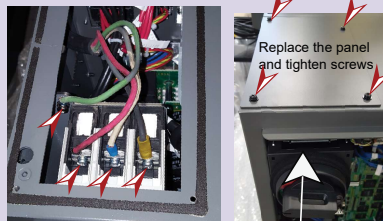
7. Route the user-supplied input power cable through the side hole. It is typical to purchase and install a strain relief cable grip (example shown below) to remove stress on the power cable connections and isolate the power cable from the controller cabinet.



8. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.



9. Verify that the input power cable is not energized. Connect it to the terminals as shown on the left below.



Insert cover

10. Re-insert the breaker cover and click it into place. Place the breaker access panel back over top of the breaker and secure with four screws.
11. With the breaker still off, energize the input power cable and use a multimeter to verify the correct voltage is present via the holes in the breaker cover.
12. Before you close the controller door, cut loose the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.



13. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.

Controller installation is complete. Go to **C. Turn on your Controller**.

C. Turn on your Controller

WARNING
Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

- Turn on the controller by rotating the Breaker clockwise to ON.
- Locate the Mode switch on the Controller Operator Panel. Use the keys retrieved in Step B12 to switch to T1 Mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.
- Pick up the teach pendant and look at the screen. Verify that the controller is in T1 Mode as seen in the figure below. If it is not, repeat the previous step and check again.
- Press the COORD key on the teach pendant until JOINT is displayed.



You have turned on your controller. You are now ready to go to **D. Jog your Robot**.

D. Jog your Robot

- If this is a new robot with no gripper or End of Arm Tooling (EOAT), you must disable the hand broken check (error **SRVO-006 Hand Broken**). To do this,
 - Press MENU, select 0 (Next), and select (SYSTEM), and then select CONFIG.
 - Use the teach pendant arrow keys to move the cursor to the Hand Broken option, and press ENTER.
 - Use the teach pendant keys to disable the hand broken setting for each motion group.

NOTE: It might already be disabled on your system.

```

System/Config
41 Signal if OVERRIDE = 100 DO[ 0]
42 Hand broken : <<GROUPS>>
43 Remote/Local setup: OP panel key
44 External I/O (ON:Remote): DI [ 0]
45 UOP auto assignment: Full
46 Multi Program Selection: TRUE
47 WAIT at Taught Position: TRUE
48 Brake control ECO mode: FALSE
49 J7,J8 jog key Setup : <<DETAIL>>
50 Collection name (F1) : [STYLE ]
51 Collection name (F2) : [COL ]
    
```

- If the error **MCTL-013 ENBL input is off** is displayed in the alarm history when you try to move the robot (it will not display as an active alarm) you must disable UI. To do this,
 - Press MENU, select 0 (Next), and select (SYSTEM), and then select CONFIG.
 - Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.
 - Use the teach pendant keys to set the option to FALSE.

```

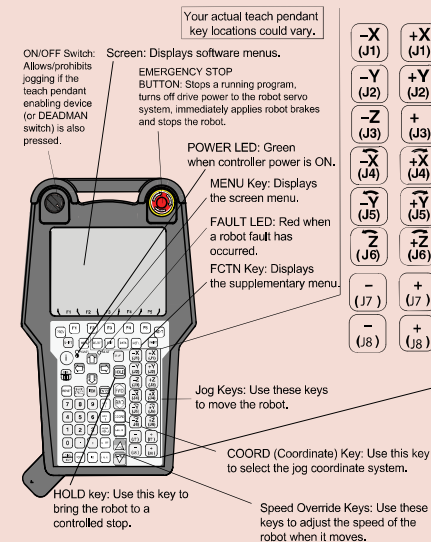
System/Config
7/59
1 Use HOT START: FALSE
2 I/O power fail recovery: RECOVER ALL
3 COLD START Autoexec program:
[*****]
4 HOT START Autoexec program:
[*****]
5 HOT START done signal: DO[ 0]
6 Restore selected program: TRUE
7 Enable UI signals: FALSE
8 START for CONTINUE only: FALSE
9 CSTOP1 for ABORT: FALSE
    
```

- Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.
- Press and hold a DEADMAN switch on the back of the teach pendant in its middle position. Press RESET on the teach pendant to clear alarms.
- To move each joint axis of the robot (*jog the robot in JOINT mode*), press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the **Robot (mechanical unit) with joints indicated** diagram on Page 1 of this guide.
- Ensure you are able to jog each robot joint.

Turning on and Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the **Controller-specific Software Installation** manual.

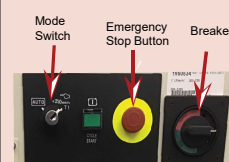
Reference

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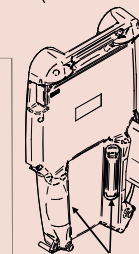


Refer to your Application-Specific Setup and Operations Manual for more information on each teach pendant key.

iPendant (Teach Pendant)



Controller Operator Panel



DEADMAN switch

For more information

To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/>.



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